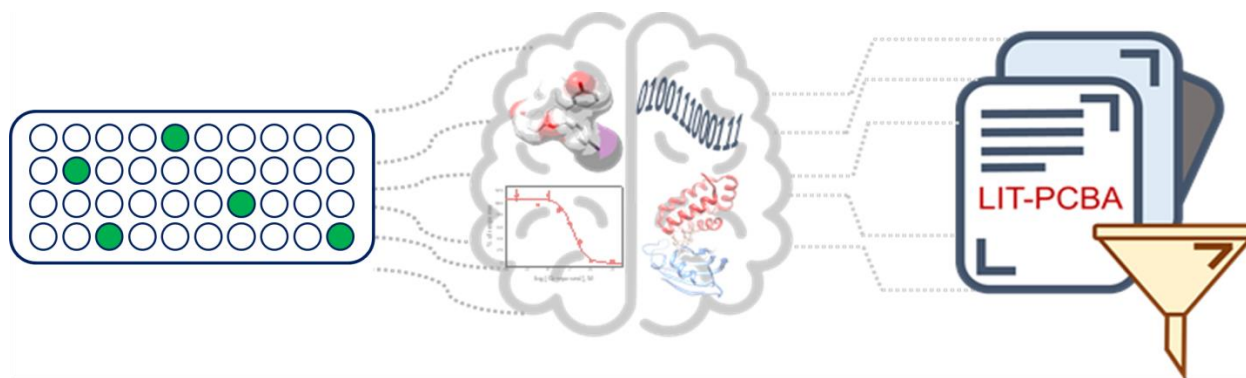


As previously mentioned in the manuscript, artificially constructed ligand sets classically used by the cheminformatics community (DUD, DUD-E, DEKOIS 2.0) suffer from multiple drawbacks ranging from the presence of possible false negatives/positives to obvious and hidden design bias, therefore overestimating the true accuracy of virtual screening methods. In this chapter, we present a novel data set entitled LIT-PCBA that was specifically designed for virtual screening and machine learning, relying on data from dose-response PubChem bioactivity assays that were additionally processed to avoid the issues inherent in other databases. The resulting ligand sets were finally unbiased by the recently described asymmetric validation embedding procedure to afford the final data collection that mimics experimental screening decks in terms of hit rate (ratio of active to **inactive** compounds) and potency distribution, and is ready for benchmarking novel virtual screening methods (both ligand-based and structure-based), notably those relying on machine learning. The work portrayed in this chapter has been published as an original research paper in the Journal of Chemical Information and Modeling, and was presented at the 9th Meeting of the French Cheminformatics Society in November 2019 in Paris.

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1. Introduction

Virtual screening (VS) of compound libraries has established itself, notably in academic settings, as a fast and cost-efficient alternative to high-throughput screening (HTS) for identifying preliminary hits of pharmaceutically interesting targets.¹⁻³ Because of the availability of hundreds of virtual screening tools,⁴ choosing the right method for a specific project often relies on benchmarking studies designed to delineate the context-specific advantages and drawbacks of each method. Many target-specific ligand sets⁵⁻¹⁰ and statistical evaluation protocols¹¹⁻¹³ have been reported during the last decade to pinpoint the ability of a VS method to prioritize, for purchase and validation, the shortest possible hit list with an optimal enrichment factor in true actives. In the early 2000s, such data sets were limited in size due to the paucity of available experimental data. **Inactive** compounds were randomly chosen from databases of drug-like chemicals.^{5,14,15} Very soon, it appeared that the random selection of presumably inactive molecules (“decoys”) led to artificially high enrichment values, because of the bias in molecular property ranges (e.g. molecular weight) that often differed between active and inactive sets.¹⁶ One of the first attempts at designing a docking-dedicated benchmarking database led to the introduction of the DUD data collection,⁶ gathering 2950 ligands of 40 different targets from the literature, seeded among property-matched decoys (36 decoys for each active) from the ZINC archive of commercially available ligands.¹⁷ In DUD, decoys were specifically designed to share physicochemical properties with actives but with a different chemical topology. Despite the caution given to the selection of decoys, independent groups rapidly noticed three major issues for both DUD active and decoy sets: (i) actives tend to spread over a few dominant scaffolds (so-called “analog bias”),¹⁸ (ii) decoys exhibited molecular net charges different from those of actives,¹⁹ and (iii) decoys were too similar to true actives and were likely false negatives.⁸ The DUD set was upgraded to a revised version (DUD-E)¹⁰ describing an enhanced and more diverse target space (102 targets), containing 22,886 clustered true actives with known experimental data from the ChEMBL database,²⁰ enhancing the proportion of decoys in the ligand sets (50 decoys for each active). The debate on the best protocol to select decoys has led to many contributions^{8,9,21} to design novel decoy sets. As an alternative to DUD-E, other sources of active compounds (e.g. PubChem BioAssay²²) have also been utilized. Note-worthy is the MUV database⁷ that provides many advantages: (i) the data sets (targets, ligands, assay conditions) are publicly available, (ii) the included compounds are drug-like, (iii) many experimental data were

utilized to remove false positives and assay artifacts, and (iv) ligands were selected by a nearest neighbor analysis to permit a spatially unbiased distribution of actives and decoys. Consequently, the MUV data collection is considered more challenging than DUD-E.²³

For many years, the DUD-E has been considered the gold standard for benchmarking VS and machine learning methods, until recent reports²⁴⁻²⁷ warned the community about both obvious and hidden bias in its design. First, Chaput *et al.*²⁴ noticed that differences in key molecular properties (polar surface area, H-bond donor count, embranchment count) remain between DUD-E actives and decoys. Moreover, chemical bias is still present in actives that tend to resemble target-bound PDB ligands, thereby overestimating the real discriminatory power of standard docking methods.²⁴ In 2018, Wallach and Heifets described the asymmetric validation embedding (AVE) method²⁵ to quantify data set bias and optimally design training/validation ligand sets. When applied to ligand-based VS methods, all standard benchmarking data collections (DUD, DUD-E, MUV) were shown to be massively biased, rewarding memorization rather than learning.²⁵ The latter danger is even higher for currently popular artificial intelligence methods (e.g. machine learning, deep neural networks)²⁸ that are hardly interpretable and tightly dependent on the quality of the input data and the way they are split to train and test a model. Two different groups^{26,27} just reported hidden bias in the DUD-E data set when applying deep neural networks (DNNs) to either predict binding affinities or classify complexes as active/inactive from X-ray structures or docking poses. Intriguingly, DNNs trained with rigorous cross-validation procedures on simple ligand descriptors were almost as accurate as those trained on protein-ligand attributes, suggesting that deep learning did not learn anything about the physics of protein-ligand interactions. Strikingly, the literature is full of overoptimistic reports describing machine learning models²⁹⁻³¹ with near perfect performances on the above-described data sets, although true VS practitioners have known for long that such an accuracy level does not mirror the proportion of experimentally confirmed hits in real prospective VS experiments.

There is more than ever an urgent need to design an unbiased and realistic data set specifically dedicated to virtual screening and machine learning.²⁷ We herewith present our contribution based on the following eight principles:

- (i) The data set should mimic “real-life” screening decks and guide VS methods to discriminate moderately potent actives (primary hits) from inactive compounds;

- (ii) The potency of all compounds (actives, inactives) for a particular target should have been determined experimentally in homogeneous conditions;
- (iii) The ratio of actives to inactives should reflect hit rates typically observed in HTS campaigns against targets of pharmaceutical interest;³²
- (iv) Actives should be filtered to remove false positives, frequent hitters, assay artifacts and truly “undruggable” compounds; besides, dose-response curves should be available for all actives;
- (v) Active and inactive compounds should span common molecular property ranges;
- (vi) Potency distribution of confirmed actives should not be biased towards too high affinities and should ideally mimic that observed in HTS decks;
- (vii) The data set should be applicable to both ligand-based and structure-based virtual screening;
- (viii) Unbiased training and validation sets should be available for machine learning.

We therefore decided to choose the PubChem BioAssay database (PCBA)²² as the source of experimental bioactivity data. PCBA is an open-access archive hosted by the National Center for Biotechnology Information (NCBI), the National Library of Medicine (NLM) and the National Institute of Health (NIH). At the time this manuscript was written, the database stores over 1 million assay records, 134,000 of which are annotated by an activity type (IC₅₀, EC₅₀, K_d, K_i). It covers about 7200 HTS projects from 80 sources (pharmaceutical companies, academic sources, governmental sources) on a chemical repository of 2.2 million compounds. The database can be easily queried according to numerous filters and is a first-class source of bioactivity data for computer-aided drug discovery.³³

We hereby describe a workflow for retrieving assays of interest and filtering compounds and targets for bioactivity data acquisition. The retrieved target sets were then subjected to state-of-the-art virtual screening experiments in order to ascertain their suitability. The final data collection entitled LIT-PCBA contains 15 targets, 7844 true active and 407,381 true inactive compounds in total; with ready-to-use input files (ligands, targets) that have been unbiased for machine learning applications. It is available for download at <http://drugdesign.unistra.fr/LIT-PCBA>.

2. Computational Methods

2.1. Data Selection

Bioactivity data were retrieved from the PubChem BioAssay database,²² where all information on true active and true inactive substances for a protein target is provided based on experimental results from confirmatory dose-response bioactivity assays, whose related details including assay principles, general protocols and other remarks are also given. All data were updated as of December 31, 2018. The “limits” search engine (<https://www.ncbi.nlm.nih.gov/pcassay/limits>) was used to filter the PubChem BioAssay resource by various options, with “Activity Outcome” set as “Active”, “Substance Type” set as “Chemical”, and “Screening Stage” defined as “Confirmatory, Dose-Response”. 149 assays, each targeting a single protein target, operated on at least 10,000 substances, and giving no fewer than 50 confirmed actives were first retained. The experimental screening data were kept if the target was characterized by at least one Protein Data Bank (PDB)³⁴ entry, in complex with a ligand of the same phenotype (i.e. inhibitor, agonist, or antagonist) as that of the tested active substances of the corresponding bioactivity assay. Altogether, 21 raw HTS data tables were directly retrieved as csv files from the PubChem BioAssay website along with actives and inactives in separate sd files. The PDB resource was then browsed by Uniprot identifiers (Uniprot IDs)³⁵ to retrieve the corresponding PDB entries in the suitable ligand-bound form.

2.2. Template Structure Preparations for Each Target Set

Protein-ligand complexes (in pdb file format) corresponding to the chosen target sets were processed as follows. For each PDB entry, explicit hydrogen atoms were added with Protoss³⁶ to any molecule (protein, cofactor, prosthetic group, ion, ligand, water). The output pdb file was then visualized in Sybyl-X 2.1.1.³⁷ A water molecule was kept under two conditions: (i) it was found at the binding site of the ligand, i.e., the distance between the oxygen atom of the water molecule and at least one heavy atom of the co-crystallized ligand was not greater than 5 Å; and (ii) it engaged in no fewer than three hydrogen bonds with the protein and/or the ligand, at least two of which were with the protein. Hydrogen bonds must satisfy the following criteria: the donor-acceptor distance must not exceed 3.5 Å; the angle formed by the donor, the hydrogen atom and the acceptor (with the vertex of the angle positioned at the hydrogen atom) must be

larger than 120 degrees. The protonated ligand and protein (including all remaining bound water molecules, cofactors, prosthetic groups and ions) were saved separately in mol2 file format with Sybyl-X 2.1.1.³⁷

In case more than 20 ligand-bound protein entries were available for each target, all protein-ligand structures were clustered according to the diversity of protein-ligand interaction patterns. These patterns were computed as graphs with IChem³⁸ as previously described,³⁹ and target-specific interaction pattern similarity matrices were computed using the GRIM score metric.³⁹ Each matrix was then used as input for agglomerative nesting clustering using the “agnes” function in R v.3.5.2, the Ward clustering method, a Euclidean distance matrix and a total number of clusters fixed to 15. For each cluster, the PDB entry with the highest resolution was chosen as the protein-ligand PDB template for the corresponding target set.

2.3. Determination of Filtering Rules for True Active and True Inactive Substances of Each Target Set

Metadata on every substance (true active and true inactive) constituting each selected target set were collected directly from the website of PubChem BioAssay, including: the substance identifier (SID), the activity label (active or inactive), the phenotype (inhibitor, agonist, or antagonist), the potency (EC₅₀ or IC₅₀, in μM), and the Hill slope for the dose-response curve of each true active. The frequency of hits (*FoH*) for a confirmed active molecule was computed as the ratio of the number of PubChem bioactivity assays in which the substance was identified as true active to the number of assays in which it was tested. Additional molecular properties (molecular weight, AlogP, total formal charge, number of rotatable bonds, number of hydrogen bond donors and acceptors) were computed in Pipeline Pilot v.19.1.0.1964.⁴⁰

For each target set, all true actives and true inactives were then filtered according to four steps:

- Step 1: Inorganic compounds filter. Molecules bearing at least one atom other than H, C, N, O, P, S, F, Cl, Br, and I were removed.
- Step 2: False positives filter (this particular step was applied only to true active substances).
 - Step 2a: Actives with Hill slope $h < 0.5$ or > 2.0 were discarded;⁷
 - Step 2b: Actives with frequency of hits $FoH > 0.26$ were removed;⁷

- Step 2c: Aggregation – auto-fluorescence – luciferase inhibition filter:⁷ actives identified as promiscuous aggregators (actives in AID 585 or AID 485341 but not in AID 584 and AID 485294), luciferase inhibitors (actives in AID 411), or compounds having auto-fluorescent properties (actives in AID 587, AID 588, AID 590, AID 591, AID 592, AID 593, AID 594) were eliminated.
- Step 3: Molecular property range filter. The remaining actives and inactives were kept if:
 - $150 < \text{molecular weight} < 800 \text{ Da}$;
 - $-3.0 < \text{AlogP} < 5.0$;
 - Number of rotatable bonds < 15 ;
 - H-bond acceptor count < 10 ;
 - H-bond donor count < 10 ;
 - $-2.0 < \text{total formal charge} < +2.0$.
- Step 4: 3D conversion and normalization filter. The two-dimensional (2D) sd files of the remaining compounds (actives, inactives) were converted into 3D sd file format using the default settings of Corina v.3.4.⁴¹ Last, compounds were standardized and ionized at physiological pH with Filter v.2.5.1.4.⁴² All preparation failures were discarded.

2.4. 2D Similarity Searches

Extended-connectivity circular ECFP4 fingerprints⁴³ were computed for PubChem compounds and PDB ligands in Pipeline Pilot v.19.1.0.1964.⁴⁰ Pairwise similarity of PubChem compounds to PDB ligands was estimated by the Tanimoto coefficient (Tc), thereby leading to a PDB ligand-specific hit list sorted by decreasing Tc values. The areas under the ROC (receiver operating characteristic)¹¹ and BEDROC (Boltzmann-enhanced discrimination of ROC)¹² curves ($\alpha = 20$) along with the enrichment in true actives at a constant 1% false positive rate over random picking (EF1%) were calculated for each separate hit list. The same procedure was applied by fusing all lists and keeping the maximal Tc value for each compound (the “max-pooling” approach).

2.5. 3D Similarity Searches

For each target set, a maximal number of 200 conformers were generated for every PubChem compound with the standard settings of Omega2 v.2.5.1.4.⁴⁴ All conformers were then compared

to the query (PDB ligand) with ROCS v.3.2.0.4.⁴⁵ The best matching conformer was selected for every ligand according to the TanimotoCombo similarity score,¹³ and all molecules of each target set were sorted based on this same value in descending order. Retrospective virtual screening performance was evaluated by ROC AUC, BEDROC AUC and EF1% values calculated as described above.

2.6. Molecular Docking

Starting from the mol2 structure of a fully processed template protein (including remaining bound water molecules after preparation) and that of its co-crystallized ligand, a protomol representing the ligand-binding site was generated from protein-bound ligand atomic coordinates using the default settings of Surflex-Dock v.3066.⁴⁶ All molecules in the relevant target set were docked into the protomol with the “-pgeom” option of the docking engine. The best-ranked pose according to docking scores (pK_d values) was retained for each molecule, and all ligands of the set were then sorted based on this value in descending order. Retrospective virtual screening performance was evaluated by ROC AUC, BEDROC AUC and EF1% values calculated as described above.

2.7. Target Set Unbiasing

For each target set, the unbiasing of the training and validation sets was done using the previously described asymmetric validation embedding (AVE) method,²⁵ which systematically measures pairwise distance in chemical space between molecules belonging to four sets of compounds (training actives, training inactives, validation actives, validation inactives). Using circular ECFP4 fingerprints⁴³ as chemical descriptors and a training-to-validation ratio of 3, a maximal number of 300 iteration steps of the AVE genetic algorithm (GA) were run to select training and validation molecules while minimizing the overall bias B ($B \in [0;1]$) of the target set. Convergence was reached when the bias value B was lower than 0.01, i.e., the GA was programmed to stop as soon as the total bias was below 0.01. To enable the script to process large sets of compounds (more than 100,000 molecules), the bias-removing script (*remove_AVE_bias.py*) originally proposed by Wallach and Heifets²⁵ was modified to allow a faster calculation on multiple cores.

3. Results and Discussion

The aim of the present study is to design an unbiased data set dedicated to virtual screening as well as machine learning, along four main ideas:

- (1) Experimental data should be available for all compounds, including the inactives. Each true active should have been confirmed by a full dose-response curve.
- (2) The target should be a single protein, for which a high-resolution X-ray structure is available on the PDB. Moreover, the target should have been crystallized at least once, with a ligand exhibiting a phenotype (e.g. inhibitor, full agonist, neutral antagonist) identical to that of active compounds in the corresponding bioassay.
- (3) PubChem target sets should be suitable for both ligand-based and structure-based virtual screening. The performance of three orthogonal methods (2D fingerprint similarity searches, 3D shape similarity searches, molecular docking) was evaluated to select only the target sets for which at least one of these three methods achieves an EF1% value ≥ 2.0 , or in other words, performs at least twice better than random picking (EF1% = 1.0).
- (4) The finally selected target sets should be as unbiased as possible, when it comes to comparing true actives to true inactives in chemical space, and when the data are split into training and validation sets.

To this end, we designed a computational workflow (**Figure 1**) that will be presented and discussed, step-by-step in the following sections.

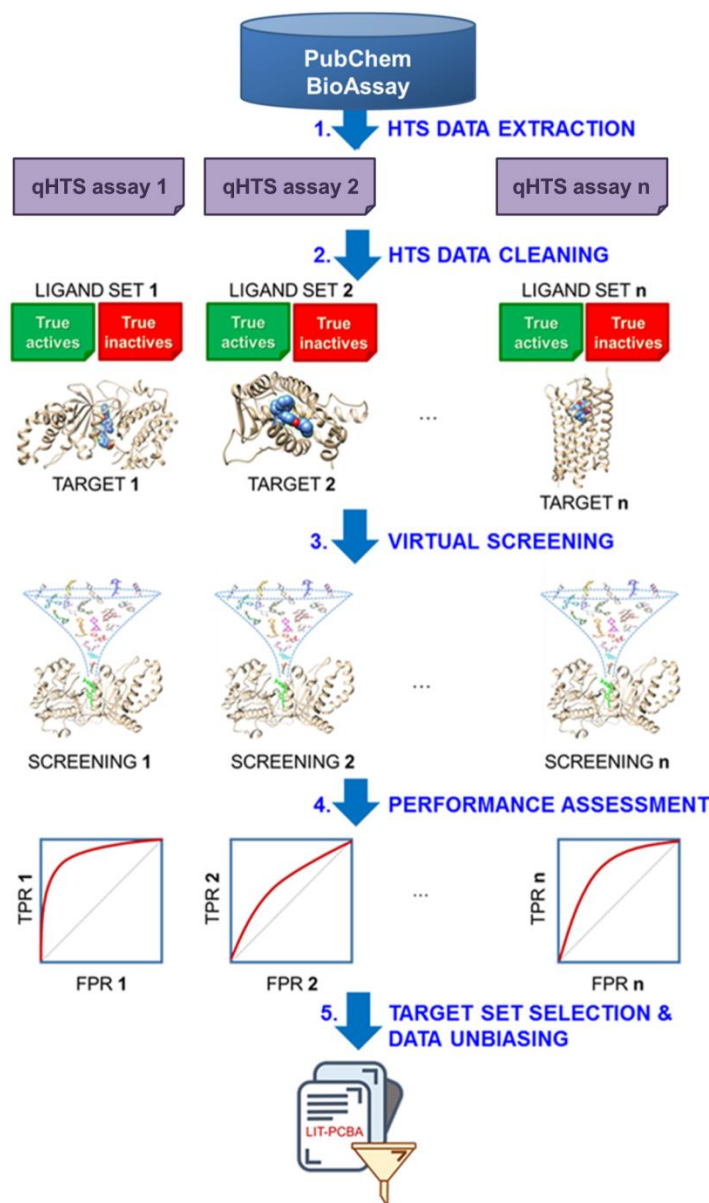


Figure 1. Workflow for LIT-PCBA data set construction. **(1)** Data retrieval from PubChem BioAssay according to user-defined filters (activity outcome: active; $\geq 10,000$ tested substances; ≥ 50 active substances; substance type: chemical; screening stage: confirmatory, dose-response; target: single; target type: protein target). **(2)** Data cleaning: removal of inorganic compounds, false positives, frequent hitters, assay artifacts and compounds with extreme molecular properties. Selection of target sets having at least a representative target structure on the Protein Data Bank co-crystallized with a ligand of the same phenotype as that of the actives in the corresponding bioassay. **(3)** Virtual screening of the cleaned HTS target sets with three methods (2D similarity, 3D similarity, docking). **(4)** Performance assessments of the methods on all cleaned target sets (ROC, BEDROC, EF1%). **(5)** Selection of target sets for which at least one method achieves an EF1% value ≥ 2.0 . AVE unbiasing²⁵ of the corresponding ligand sets and definition of training and validation sets for machine learning.

3.1. HTS Data Extraction

PubChem (<https://pubchem.ncbi.nlm.nih.gov>) is a public repository for information on 91 million chemical substances and 268 million biological activities, launched in 2004 as a component of the Molecular Libraries Roadmap Initiatives of the US National Institutes of Health (NIH). The PubChem BioAssay resource²² was queried to retrieve 149 assays according to multiple queries (see “Computational methods”). To ascertain that the data set will be further suitable for both ligand-based and structure-based virtual screening, we checked that each single protein target not only had a representative structure on the PDB, but was also co-crystallized with a ligand sharing the same phenotype or function with the true actives. This sanity check enables the selection of the right activation state (e.g. for G-protein coupled receptors) and the right binding site for docking. Of course, we cannot ensure at this step that all true actives share the same binding site with all PDB ligand templates. However, it serves as the first filter to avoid comparing ligands with known opposite or different functions. To control the bioactivity of each compound, only confirmatory dose-response screening assays were kept. A total of 21 assays (**Table 1**) performed on isolated enzymes ($n = 6$), soluble protein-protein interactions ($n = 4$) and target-expressing cells ($n = 11$); using four different readouts (fluorescence intensity, fluorescence polarization, luminescence, alpha screen) were finally saved. Except for five screens in which only 10,000 compounds were tested, most assays were run on a large number of compounds (from 200,000 to 400,000). Importantly, each assay was analyzed in detail, notably regarding the activity threshold qualifying a compound as active, which is target-dependent and was not further modified in this study. Compounds whose activity outcome was deemed as “inconclusive” were removed from the final ligand sets, only ligands confirmed as either actives or inactives were retained.

Corresponding targets are single proteins representing 11 families of pharmaceutical interest, including nuclear hormone receptors ($n = 5$), protein kinases ($n = 3$), and G protein-coupled receptors ($n = 3$). Most target sets describe compounds tested for an inhibitory activity against a protein target (13 target sets). Overall, 162 structures of protein-ligand complexes in PDB format were chosen as templates for the 21 target sets (**Table 1**). More information on each selected PubChem bioactivity assay (brief assay description, readout, format, PDB templates) can be found in **Table S1**.

Table 1. List of 21 selected PubChem bioactivity assays

ID	Target	Assay AID ^a	Tested	Substances ^b		PDB entries
	Name			Actives	Phenotype	
ADRB2	Beta2 adrenergic receptor	492947	331,108	80	Agonist	8
ALDH1	Aldehyde dehydrogenase 1	1030	220,402	16,117	Inhibitor	8
ARO1	Aromatase	743083	10,486	905	Inhibitor	3
ESR1-ago	Estrogen receptor alpha	743075	10,486	589	Agonist	15
ESR1-ant	Estrogen receptor alpha	743080	10,486	477	Antagonist	15
FEN1	Flap endonuclease 1	588795	391,275	1368	Inhibitor	1
GBA	Glucocerebrosidase	2101	326,770	299	Inhibitor	6
GLP1R	Glucagon-like peptide-1 receptor	624417	408,352	6432	Inverse agonist	2
GLS	Glutaminase	624170	409,400	846	Inhibitor	11
IDH1	Isocitrate dehydrogenase	602179	390,606	365	Inhibitor	14
KAT2A	Histone acetyltransferase KAT2A	504327	387,485	817	Inhibitor	3
L3MBTL1	Lethal(3)malignant brain tumor-like protein isoform I	485360	225,505	1495	Inhibitor	1
MAPK1	Mitogen-activated protein kinase 1	995	72,004	711	Inhibitor	15
MTORC1	Mechanistic target of rapamycin	493208	43,989	342	Inhibitor	11
OPRK1	Kappa opioid receptor	1777	284,220	51	Agonist	1
PKM2	Pyruvate kinase muscle isoform 2	1631	264,516	892	Agonist	9
PPARG	Peroxisome proliferator- activated receptor gamma	743094	10,486	78	Agonist	15
RORC	Retinoic acid-related orphan receptor gamma	2551	309,031	16,824	Inhibitor	15
THRB	Thyroid hormone receptor	1469	282,587	183	Inhibitor	1
TP53	Cellular tumor antigen p53	651631	10,488	602	Agonist	6
VDR	Vitamin D receptor	504847	401,452	3735	Antagonist	2

^a Full details for each assay are available at <https://pubchem.ncbi.nlm.nih.gov/bioassay/AID>.

^b Structures deposited by individual data contributors. Unique chemical structures are called “compounds”.

3.2. HTS Data Cleaning

All active and inactive compounds were next submitted to a series of filters (see “Computational methods”) aimed at removing inorganic compounds (step 1), frequent hitters and assay artifacts (step 2),⁷ compounds exhibiting molecular properties outside pre-defined ranges (step 3), and molecules for which either 2D-to-3D conversion or ionization at pH 7.4 failed (step 4). It can be observed that nearly 60% of true active substances were removed during the filtering steps (see **Table S2, S3** for exhaustive statistics), with step 2a eliminating the most true actives (**Figure 2**). This step is aimed at ruling out actives that exhibit very strong binding cooperativity and have multiple binding sites.⁴⁷ True inactive substances, on the other hand, were not subjected to the three filtering steps 2a, 2b and 2c, thus lost much fewer members than the true actives, with over 90% of substances still remaining in the end.

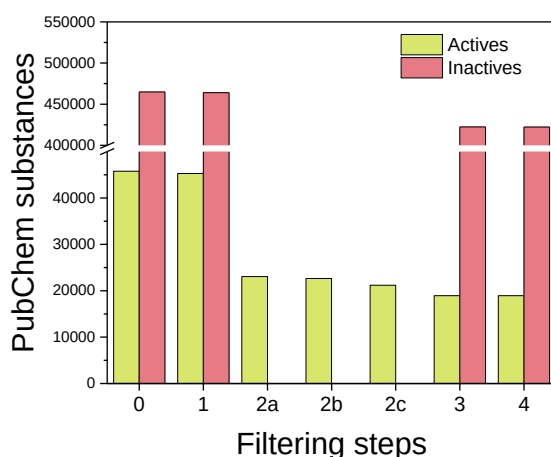


Figure 2. Total number of actives and inactives remaining after each filtering step was applied to the 21 selected target sets from PubChem BioAssay. Step 1: inorganic molecules; Step 2a: compounds with Hill slope $h < 0.5$ or > 2 ; Step 2b: compounds with frequency of hits $FoH > 0.26$; Step 2c: assay artifacts interfering with the readouts (10,892 substances classified as aggregators or auto-fluorescent molecules or luciferase inhibitors); Step 3: compounds with extreme molecular properties; Step 4: 3D conversion and ionization failures. Steps 2a, 2b and 2c were not applied to true inactives.

The filtering steps highlight the importance of removing assay artifacts in the composition of active substances. These steps not only prevented false positives that could affect subsequent screening performance, but also significantly reduced the number of true actives in comparison to that of true inactives, thus bringing hit rates closer to those typically observed in experimental

screening decks,³² but lower (in 15 out of 21 cases) than those of artificially constructed data sets commonly used by the cheminformatics community (**Figure 3A**).

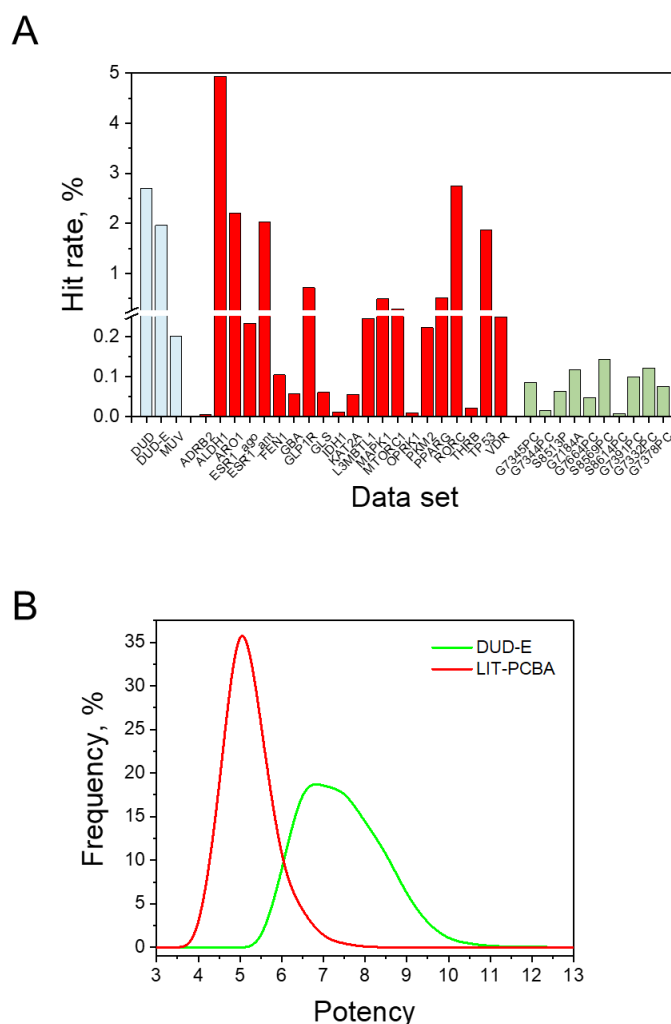


Figure 3. Properties of LIT-PCBA and standard data sets. **(A)** Confirmed hit rates for the LIT-PCBA data set (red bars), standard cheminformatics data sets (DUD,⁶ DUD-E,¹⁰ MUV;⁷ blue bars), and a representative sample of 10 high-throughput screens from a major pharmaceutical company (green).³² **(B)** Potency distribution of actives in the LIT-PCBA (red) and DUD-E (green) data sets. Potency is expressed as pIC₅₀, pEC₅₀, pK_i or pK_d.

We next looked at the potency distribution of true actives (**Figure 3B**) in our data set in comparison to that of DUD-E and ChEMBL.²⁰ We can observe different potency distribution for DUD-E actives ($n = 67,659$; median potency = 7.46 ± 0.96) and for LIT-PCBA actives ($n = 19,985$; median potency = 5.22 ± 0.54). The micromolar potency values observed for most LIT-PCBA actives reflect affinities typically observed in HTS campaigns. Conversely, DUD-E

actives tend to be much more potent (potency at the sub-micromolar level in most cases) and consequently easier to be picked, thereby overestimating the real benefit of virtual screening methods. At the individual target set level, the same trend applies when comparing the potency of LIT-PCBA and ChEMBL ligands for 19 common target sets (**Figure S1**). Importantly, we believe that the enhanced difficulty proposed by our data collection may enable a better discrimination of *in silico* screening methods.

3.3. Virtual Screening and Performance Assessments

The suitability of the 21 fully processed target sets for virtual screening was next assessed by three standard methods: 2D fingerprint similarity searches, 3D shape similarity searches, and molecular docking. The aim of the computational experiments was not to compare the virtual screening accuracy degrees of all methods but to check which of the 21 target sets may be unsuitable for *in silico* screening purposes. Hence, there is no guaranty that PubChem and PDB template ligands are strictly comparable (e.g. sharing the same binding site and molecular mechanism of action). Ligand-based screening will rapidly assess whether obvious bias is present in the ligand sets in terms of either 2D or 3D topologies. In addition, docking will ascertain if PubChem ligands share binding sites and interaction patterns with PDB templates. In each screen, all available PDB ligand/target templates were iteratively used as references, thereby generating as many hit lists as the available 162 templates. This exhaustive approach, albeit cumbersome, enables the selection of all references and takes into account the known chemical diversity of target-bound ligands (ligand-based virtual screening) or the known conformational space accessible to the target of interest (docking). In addition, a target-based “max-pooling” approach was followed by merging all screening data related to any LIT-PCBA ligand, whatever the corresponding template, and retaining the highest value (2D similarity, 3D similarity, docking score) per ligand. Statistical analyses of the data were primarily focused on enrichment factors in true actives at a constant 1% false positive rate (EF1%, **Figure 4**) as such values mirror the expectation of prospective virtual screening practices. Besides, areas under the ROC and BEDROC curves have also been calculated and are given in **Tables S4-S6**.

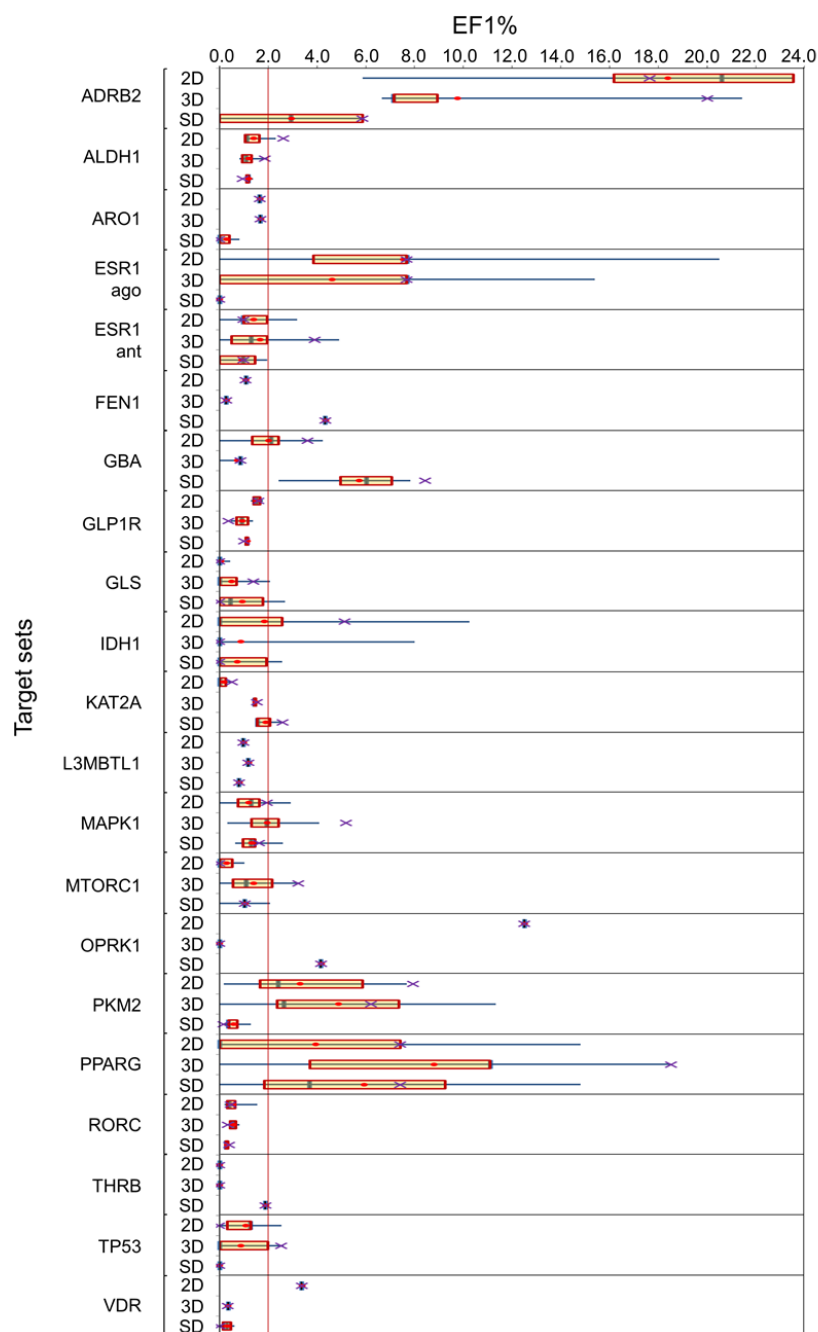


Figure 4. Performance of three different virtual screening methods (2D: ECFP4 similarity, 3D: 3D shape similarity, SD: molecular docking with Surflex-Dock) on 21 fully processed target sets. The graphs represent the distribution of EF1% values (enrichment in true actives at a constant 1% false positive rate over random picking) obtained after screening. The boxes delimit the 1st and 3rd quartiles, and the whiskers delimit the minimum and the maximum values. The median and the mean values are indicated by a green vertical line and a red dot located in each box, respectively. In cases where there is only one PDB template for a target set, or all templates gave the same EF1% value, the boxes are shrunk down into a single line. The purple crosses represent the EF1% values obtained by the “max-pooling” approach.

As expected, inspection of the observed enrichment in true actives for all 21 target sets clearly shows that the EF1% values may vary quite significantly according to the chosen template. In many instances, enrichment close to or even poorer than that obtained by random picking (EF1% = 1.0) is observed (**Figure 4**). We considered as acceptable any virtual screening protocol yielding an EF1% value ≥ 2 , or in other words, at least twice better than random picking. At this threshold, ligand-based methods clearly outperformed docking (**Figure 4**). Interestingly, only 10% of all *in silico* screening assays led to enrichment higher than 10. This result highlights the particular challenge of screening the current data set that we attribute to two main reasons: (i) the apparent absence of obvious bias in the distribution of PubChem actives in chemical space, and (ii) the potency distribution of PubChem actives not centered on sub-micromolar values.

3.4. Final Target Set Selection and Unbiasing

In order to facilitate the analysis, we will from now on discuss the results obtained by fusing, for each virtual screening method, all data across all available target-specific templates (“max-pooling” approach). This strategy was supported by two main reasons: (i) the fused approach provides enrichment values usually close to that obtained with the best possible template (**Figure 4**), and (ii) it enables the definition of a single hit list for each screening run while considering all templates. 15 out of the initial 21 target sets can be considered suitable (EF1% ≥ 2.0) for at least one of the three methods (**Figure 5**).

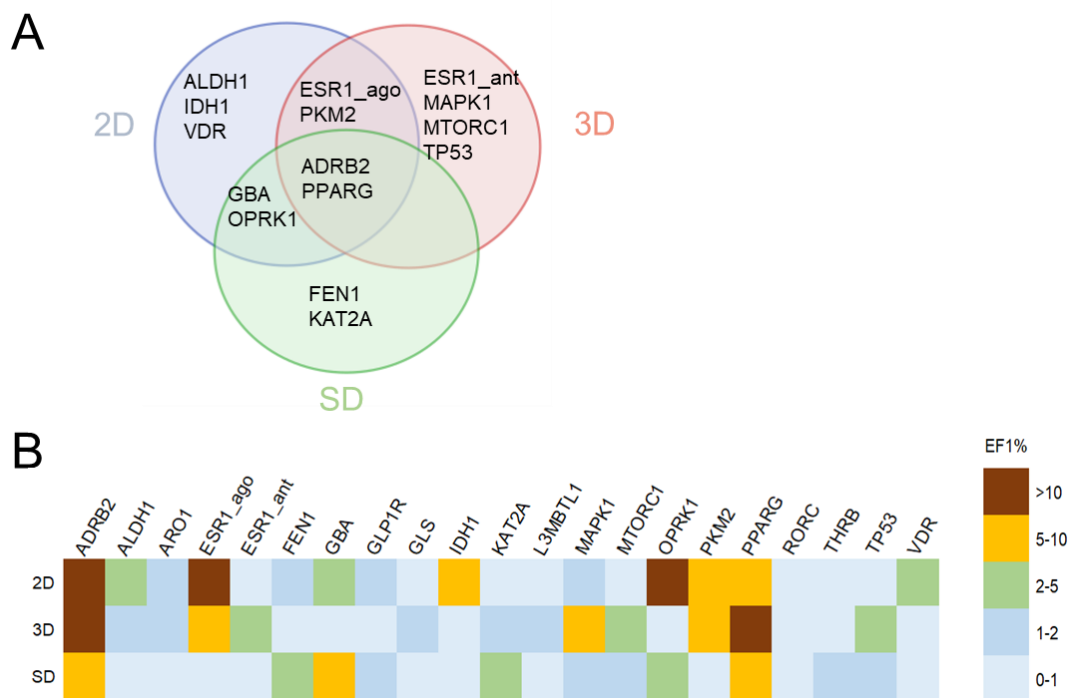


Figure 5. Comparative performance of three virtual screening protocols (2D: ECFP4 similarity searches, 3D: shape similarity searches, SD: molecular docking with Surflex-Dock) for the 21 target sets, processed by the “max-pooling” approach. **(A)** Venn diagram of target sets for which an EF1% ≥ 2.0 is observed. **(B)** Heatmap representing the fused values of EF1% obtained for each of the 21 fully processed target sets by the three *in silico* screening methods. Abbreviations of target sets are indicated above the heat map.

The current virtual screening exercise suggests that six target sets (GLS, GLP1R, ARO1, THRB, RORC, L3MBTL1) are not adequate for *in silico* screening purposes since none of the three methods was able to clearly distinguish the confirmed actives from inactive compounds when the “max-pooling” approach was applied (EF1% < 2.0) (**Figures 4-5**). Moreover, for five target sets among them (GLS as the only exception), the template-based scoring approach did not give any EF1% value above 2.0 either. Reasons for failures in screening these targets were: (i) the promiscuity of the binding site towards many low-affinity chemotypes (e.g. ARO1), (ii) the presence of non-overlapping binding sites (orthosteric versus allosteric) for PDB templates and PubChem actives (e.g. GLP1R, GLS, RORC), and (iii) the availability of a single PDB template (e.g. L3MBTL1, THRB).

Two target sets (ADRB2, PPARG) seem easier to handle since all three virtual screening methods could successfully retrieve true actives with enrichment factors higher than 5.0. In four

cases (GBA, OPRK1, PKM2, ESR1-ago), two methods succeeded. Last, only one method was able to perform correctly for 9 sets (ALDH, IDH1, VDR, MTORC1, MAPK1, ESR1-ant, TP53, FEN1, KAT2A; **Figure 5**). This result is in agreement with many previous studies⁴⁸⁻⁵⁰ suggesting that *in silico* screening methodologies are orthogonal, and is reassuring as it highlights the absence of obvious bias in both 2D molecular graphs and 3D shapes of LIT-PCBA compounds. It can therefore be implied that the remaining true actives (besides the ADRB2 and PPARG sets) do not resemble their corresponding PDB template ligands in both 2D and 3D shapes; meaning similarities between them, if there were any, did not significantly contribute to improving virtual screening performance, notably in early enrichment of true actives.

For each of the remaining 15 target sets, we ensured that the chemical diversity of PDB template ligands was not biasing our analysis. A first comparison of the number of Bemis-Murcko frameworks⁵¹ to the total number of templates indicates that a wide variety of chemotypes are indeed available among the chosen PDB template ligands (**Table S7**). A self-similarity plot of templates (Tanimoto coefficient on MDL public keys) confirms this observation and shows, for most of the target sets (MTORC1 being an exception), a large chemical diversity (**Figure S2**).

The 15 target sets were last unbiased by the AVE method²⁵ to propose optimal training and validation sets for machine learning applications. In brief, a genetic algorithm (GA) is used to select four subsets of active and inactive compounds for training and validation sets, based on pairwise distances in chemical space (ECFP4 circular fingerprints) between the above-described four ligand subsets. The objective function of the GA (bias value) gears the splitting procedure to select training and validation sets for which distances in chemical space are homogeneously distributed when training actives, validation actives, training inactives and validation inactives are compared. For 14 out of 15 target sets, just a few iterations (< 100) of the GA were necessary to unbiased the corresponding target sets with low bias values (**Table 2**). Interestingly, optimal splitting was achieved without removing a single compound from 13 out of the 15 initial PubChem compound collections, thereby suggesting that the latter input did not exhibit major bias. The final AVE-unbiased LIT-PCBA data set covers 15 target sets, 7844 unique actives and 407,381 unique inactives (**Table 2**).

For two target sets (ALDH1, VDR), the high number of true actives forced us to reduce by 25% the size of the data set in order to reach GA search completion. In both cases, care was taken to

keep the hit rates unchanged after data reduction. A one-nearest neighbor (knn1) binary classification of the 15 validation sets, still using ECFP4 fingerprints as descriptors, led to areas under the ROC curves close to random (0.50) and thereby supports the *bona fide* unbiasing of all corresponding target sets. Analyses of the three baseline virtual screening experiments for the AVE validation sets only (**Table S8**) confirm the very challenging nature of the data set as the performance dropped drastically for many target sets, notably those with low numbers of actives (e.g. ADRB2, IDH1) or few PDB template ligands (e.g. OPRK1). As previously indicated, the baseline *in silico* screening protocol was just intended to remove PubChem HTS data unsuitable for virtual screening applications, and is not indicative of the performance of modern machine learning approaches. We however recommend the application of such methods to target sets exhibiting enough true actives to train on (ALDH1, FEN1, GBA, KAT2A, MAPK1, PKM2, VDR; **Table 2**).

Table 2. Final list of 15 target sets of the LIT-PCBA data collection

Target	Target name	AVE		Actives		Inactives		Knn1 ^a ROC AUC
		Bias	Iterations	Validation	Training	Validation	Training	
ADRB2	Beta2 adrenergic receptor	0.003	2	4	13	78,120	234,363	0.500
ALDH1 ^b	Aldehyde dehydrogenase 1	0.092	195	1344	4032	25,868	77,606	0.556
ESR1-ago	Estrogen receptor alpha	0.001	1	3	10	1395	4188	0.499
ESR1-ant	Estrogen receptor alpha	0.006	9	25	77	1237	3711	0.517
FEN1	Flap endonuclease 1	0.076	39	92	277	88,850	266,552	0.499
GBA	Glucocerebrosidase	0.005	9	41	125	74,013	222,039	0.524
IDH1	Isocitrate dehydrogenase	0.001	4	9	30	90,512	271,537	0.500
KAT2A	Histone acetyltransferase KAT2A	0.001	5	48	146	87,137	261,411	0.500
MAPK1	Mitogen-activated protein kinase 1	0.000	8	77	231	15,657	46,972	0.505
MTORC1	Mechanistic target of rapamycin	0.001	7	24	73	8243	24,729	0.499
OPRK1	Kappa opioid receptor	0.000	3	6	18	67,454	202,362	0.500
PKM2	Pyruvate kinase muscle isoform 2	0.009	28	136	410	61,380	184,143	0.507
PPARG	Peroxisome proliferator-activated receptor γ	0.000	4	6	21	1302	3909	0.500
TP53	Cellular tumor antigen p53	0.008	29	19	60	1042	3126	0.491
VDR ^b	Vitamin D receptor	0.044	62	165	498	66,635	199,906	0.499

^a Area under the ROC curve for a binary classification of validation compounds (active, inactive) based on a one-nearest neighbor similarity search (ECFP4 fingerprints) model trained on target-specific training sets.

^b The size of the target set was reduced by 25% at the unbiasing stage due to the large number of remaining true actives.

4. Conclusion

A rigorous ligand set preparation process is necessary to benchmark virtual screening and/or machine learning methods. Since the body of known experimental data is continuously increasing, such benchmarking data sets need periodical revisions to remove both obvious and hidden bias inherent in human decision-making. Otherwise, errors are propagated across the literature and prevent a true comparison of novel methodological developments. Several recent reports²⁴⁻²⁷ unambiguously demonstrated that the cheminformatics community is currently facing this situation, leading notably to overoptimistic reports on the real benefit of artificial intelligence methods (e.g. deep neural networks) when applied to structure-based ligand design. We herewith present LIT-PCBA as a novel generation of virtual screening benchmarking data sets, specifically designed to reveal the true potential of computational methods in *in silico* screening exercises. The data collection has been designed from dose-response PubChem bioactivity assays for which active and inactive compounds are unambiguously defined. Importantly, a careful examination of metadata allowed the removal of assay artifacts, frequent hitters and false positives. LIT-PCBA comprises 15 target sets covering a wide diversity of ligands and target proteins. Preliminary virtual screening attempts with state-of-the-art methods (2D similarity searches, 3D shape-matching, and molecular docking) suggest that the data set is very challenging, notably because potency distribution bias among the labeled active compounds is no longer present. A recently described unbiasing procedure²⁵ was finally applied to LIT-PCBA to enable a rational and optimal distribution of training and validation sets for machine learning. We do believe that the particular challenge brought by this data collection will allow a clearer appreciation of modern artificial intelligence methods in structure-based virtual screening scenarios. The full LIT-PCBA data set is now freely accessible for download at <http://drugdesign.unistra.fr/LIT-PCBA>.

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